

## Ideation Phase

### Define the Problem Statements

|               |                                                                   |
|---------------|-------------------------------------------------------------------|
| Date          | 18 July 2026                                                      |
| Team ID       | SWTID-2026-7520                                                   |
| Project Name  | Visualization Tool for Electric Vehicle Charge and Range Analysis |
| Maximum Marks | 2 Marks                                                           |

#### Customer Problem Statement Template:

Electric vehicle (EV) users often struggle to understand how their battery charge, driving behavior, and road conditions impact their vehicle's range. Current EV systems provide only basic battery level information without detailed visual insights or predictive analysis. This lack of clarity leads to range anxiety, inefficient energy usage, and uncertainty during trip planning. Customers need a smart, easy-to-use visualization tool that helps them monitor charge trends, predict range accurately, and make informed driving decisions with confidence.

